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MOODTUNE AI: SONG EMOTION ANALYSIS AND EMOTION-BASED MUSIC RECOMMENDATION USING DISTILBERT

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MINI PROJECT REPORT

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CHAPTER NO.	TABLE OF CONTENTS	PAGE NO.
	ABSTRACT	**i**
	LIST OF FIGURES	**ii**
	ABBREVIATIONS	**iii**
1	**INTRODUCTION**	**1**
	1.1 Overview of the Project	1
	1.2 Problem Statement	1
	1.3 Objectives of the Project	2
	1.4 Scope of the Project	2
	1.5 Organization of the Report	3
2	**LITERATURE REVIEW**	**4**
	2.1 General	4
	2.2 Existing System	4
	2.2.1 Literature Survey on Emotion Classification	4
	2.2.2 Literature Survey on Transformer-Based Models	5
	2.3 Proposed System	5
3	**SYSTEM DESIGN**	**7**
	3.1 General	7
	3.2 System Architecture	7
	3.3 Module Description	8
	3.3.1 Data Collection Module	8
	3.3.2 Data Preprocessing Module	8
	3.3.3 Automated Emotion Labeling Module	8
	3.3.4 Class Imbalance Handling Module	9
	3.3.5 Model Training Module	9
	3.3.6 Model Evaluation Module	9
	3.3.7 Web Application Module	9
	3.3.8 Recommendation Module	9
	3.4 Data Flow	10
	3.5 Dataset Design	11
	3.6 Software Requirements	11
	3.7 Design Limitations	12
4	**IMPLEMENTATION**	**13**
	4.1 Methodology	13
	4.2 Data Collection and Preprocessing	13
	4.2.1 Class Distribution	13

CHAPTER NO.	TABLE OF CONTENTS	PAGE NO.
	4.3 Automated Emotion Labeling	14
	4.4 Class Imbalance Handling	14
	4.5 Model Training	15
	4.6 Model Evaluation	15
	4.7 Demonstration of the Web Application	18
	4.8 Important Code Implementation	22
	4.8.1 Flask Request Handling	22
	4.8.2 DistilBERT Emotion Prediction	23
	4.8.3 Emotion-Based Song Recommendation	24
	4.9 Chapter Summary	25
5	**RESULTS AND DISCUSSION**	**26**
	5.1 Introduction	26
	5.2 Model Performance Evaluation	26
	5.3 Comparison of Experimental Approaches	26
	5.4 Confusion Matrix and Class-Wise Results	27
	5.5 Web Application Results	28
	5.6 Limitations of the Results	28
	5.7 Chapter Summary	29
6	**CONCLUSION AND FUTURE ENHANCEMENTS**	**30**
	6.1 Conclusion	30
	6.2 Future Enhancements	30
	REFERENCES	**32**

ABSTRACT

Music is an important medium for expressing emotions, and users often prefer songs that match their emotional state. Most existing music recommendation systems rely mainly on listening history, genre, artist preference, or popularity, and give limited importance to the emotional meaning of song lyrics. This project presents MoodTune AI, a Flask-based system that uses Natural Language Processing (NLP) and a fine-tuned DistilBERT model to classify song lyrics into six emotion categories: Anger, Fear, Happy, Love, Motivational, and Sad. The user enters a song title; the system retrieves the lyrics from a local dataset, predicts the emotion with a confidence score, and recommends up to five songs from the same emotion category. Four training approaches were compared: baseline training, resampling, class-weighted loss, and dataset expansion. The final model, trained on an expanded dataset of 5,941 labeled songs, achieved a reported accuracy of 67.6% (804 of 1,189 test samples) and a Motivational-class F1-score of 0.52. The baseline reached a higher accuracy of 72% but obtained a Motivational F1-score of 0.00. The results are limited by automatically generated labels, class imbalance, and dependence on the lyrics available in the local dataset. Future work includes manually verified labels, multilingual lyric analysis, audio features, personalized recommendation, and integration with music streaming platforms.

Keywords: Natural Language Processing (NLP); DistilBERT; Song Emotion Analysis; Music Recommendation; Emotion Classification; Artificial Intelligence.

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
Figure 3.1	System Architecture of MoodTune AI	8
Figure 3.2	Modules of MoodTune AI	10
Figure 4.1	Distribution of Emotion Classes in the Dataset	14
Figure 4.2	Comparison of Accuracy and Motivational F1-Score Across Experiments	16
Figure 4.3	Per-Class F1-Score Comparison	16
Figure 4.4	Confusion Matrix of the Final DistilBERT Model	17
Figure 4.5	Web Application Output for the Anger Example (“Dangerous Woman”)	18
Figure 4.6	Web Application Output for the Fear Example (“bury a friend”)	19
Figure 4.7	Web Application Output for the Happy Example (“Firework”)	19
Figure 4.8	Web Application Output for the Love Example (“Heartbeat”)	20
Figure 4.9	Web Application Output for the Motivational Example (“Stronger”)	20
Figure 4.10	Web Application Output for the Sad Example (“Fix You”)	21

ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma-Separated Values
DistilBERT	A smaller, distilled transformer language model based on BERT
F1-score	Harmonic mean of precision and recall
Flask	Python web application framework used by the project
HTTP	Hypertext Transfer Protocol
ML	Machine Learning
NLP	Natural Language Processing
POST	HTTP request method used to submit form data
Softmax	Function that converts logits into class probabilities
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1. INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

Music is an important part of everyday life and has a strong connection with human emotions. People listen to music to relax, improve their mood, express their feelings, and find motivation. Different songs can express different emotions, such as happiness, sadness, love, fear, anger, and motivation. However, identifying the emotional content of a song and finding similar songs manually can be difficult when dealing with large collections of music.

MoodTune AI is an intelligent music recommendation system that uses Natural Language Processing (NLP) and Artificial Intelligence (AI) to analyze song lyrics and recommend songs based on their emotional content. The system uses a fine-tuned DistilBERT transformer model to classify song lyrics into six emotional categories: Anger, Fear, Happy, Love, Motivational, and Sad.

The application is developed using Python and Flask. Users can enter a song title through a simple web interface. The system searches for the song in its local dataset, retrieves the corresponding lyrics, and processes them using the trained DistilBERT model. Based on the lyrics, the model predicts the emotion and displays the prediction confidence. The system then recommends up to five songs from the same emotional category and provides YouTube search links for easy access.

The project also investigates different approaches to improve emotion classification, including baseline training, resampling, class-weighted loss, and dataset expansion. The final reported model achieves an accuracy of 67.6%, with a Motivational-class F1-score of 0.52. These results highlight the challenges involved in classifying song emotions, particularly when some categories have fewer training examples.

MoodTune AI demonstrates how transformer-based language models can be applied to music discovery and emotion-aware recommendations. The system provides a foundation for developing more advanced music recommendation applications that consider the emotional meaning of song lyrics.

1.2 PROBLEM STATEMENT

Traditional music recommendation systems often depend on listening history, popularity, artist preferences, or genre. Although these methods are useful, they may not directly consider the

emotional meaning expressed in song lyrics. Users who want songs matching a particular feeling may need to search manually through multiple songs and playlists.

Automatically identifying emotions from song lyrics is also challenging because lyrics may contain figurative language, ambiguous expressions, and overlapping emotional meanings. In addition, an imbalanced dataset can make it difficult for a classification model to recognize less-represented emotions accurately.

Therefore, the project aims to develop a web-based system that predicts the emotion of a song from its lyrics and recommends other songs belonging to the same emotional category.

1.3 OBJECTIVES OF THE PROJECT

The main objectives of MoodTune AI are:

1. To develop a dataset containing song titles, lyrics, and emotion labels.
2. To preprocess song lyrics for use in a Natural Language Processing model.
3. To apply a fine-tuned DistilBERT model for classifying song lyrics into six emotional categories.
4. To investigate methods such as class-weighted loss and dataset expansion to improve classification performance.
5. To develop a user-friendly web application using Flask.
6. To display the predicted emotion and its confidence score for a selected song.
7. To recommend up to five songs with the same predicted emotion.
8. To provide YouTube search links that help users access the recommended songs.

1.4 SCOPE OF THE PROJECT

The scope of MoodTune AI is limited to emotion classification based on song lyrics available in the project's local dataset. The application accepts a song title, searches the dataset for the corresponding lyrics, and uses the trained model to predict one of six emotional categories: Anger, Fear, Happy, Love, Motivational, and Sad.

The system provides a web interface for displaying the predicted emotion, confidence score, and songs with matching emotion labels. It also provides YouTube search links for the recommended songs.

The current system does not perform real-time audio analysis, detect a user's facial expressions, or directly measure the listener's emotional state. Recommendations are based on the predicted emotion of the selected song rather than individual listening history or personal preferences. The system's performance also depends on the quality, coverage, and balance of the available dataset.

1.5 ORGANIZATION OF THE REPORT

This report is organized into chapters that explain the development of MoodTune AI.

Chapter 1 – Introduction: Presents the project overview, problem statement, objectives, and scope.

Chapter 2 – Literature Review: Discusses existing research on music emotion classification, Natural Language Processing, and transformer-based models.

Chapter 3 – System Design: Explains the proposed architecture, system modules, data flow, dataset design, and software requirements.

Chapter 4 – Implementation: Describes the methodology, data preprocessing, emotion labeling, model training, evaluation, Flask application, and recommendation process.

Chapter 5 – Results and Discussion: Presents the reported model performance, evaluation results, sample predictions, and limitations.

Chapter 6 – Conclusion and Future Enhancements: Summarizes the project and identifies possible improvements and future extensions.

CHAPTER 2. LITERATURE REVIEW

2.1 GENERAL

Music emotion recognition is an important research area that combines Artificial Intelligence, Natural Language Processing, and music information retrieval. The main objective is to identify the emotions expressed by music and use this information to improve music discovery and recommendation. Song lyrics provide valuable information about emotional meaning through words, phrases, and sentences.

Traditional text classification techniques often depend on manually designed features and statistical representations of words. These methods can identify basic patterns in text, but they may have difficulty understanding context and complex emotional expressions. Recent advances in transformer-based language models have improved the ability of computers to understand the contextual meaning of natural language.

MoodTune AI applies a transformer-based approach to classify song lyrics into six emotional categories and recommend songs with matching emotion labels.

2.2 EXISTING SYSTEM

Existing music recommendation systems commonly use listening history, song popularity, artist preferences, genre, and user ratings to recommend music. These approaches can provide relevant suggestions, but they may not directly analyze the emotional meaning of song lyrics.

Some emotion recognition approaches use manually selected keywords or traditional machine learning algorithms. Although these techniques can be useful, their performance may be affected by ambiguous words, context-dependent meanings, and variations in how emotions are expressed.

Another challenge is class imbalance. If certain emotions appear less frequently in the training dataset, the model may learn to recognize common emotions more effectively than rare ones. This can reduce the reliability of recommendations for underrepresented categories.

2.2.1 LITERATURE SURVEY ON EMOTION CLASSIFICATION

Research in music emotion recognition has explored different ways of identifying emotional characteristics from musical content. Some approaches analyze audio features, while others use lyrics to identify emotions expressed through language. Lyrics-based approaches are particularly relevant when the objective is to understand the textual meaning of a song.

Traditional Natural Language Processing methods use techniques such as word frequency, TF-IDF, and machine learning classifiers to represent and classify text. These methods provide a useful foundation for text classification, but they may not fully capture the contextual meaning of words and sentences.

Research on lyric-based music recommendation demonstrates the importance of connecting textual emotion analysis with recommendation techniques. However, the quality of the resulting recommendations depends on the reliability of the emotion labels and the coverage of the song dataset.

2.2.2 LITERATURE SURVEY ON TRANSFORMER-BASED MODELS

Transformer models use attention mechanisms to learn relationships between words in a text sequence. Unlike approaches that mainly rely on individual words, transformer-based models can represent contextual information and help identify emotional meaning in lyrics. Transformer-based models have also been applied to emotion classification in general text [9].

BERT is a widely used transformer architecture for language understanding tasks [15]. DistilBERT is a smaller, more computationally efficient model derived from BERT [14]. It retains many of the contextual language-understanding capabilities of the original model while reducing model size and computational requirements. DistilBERT has about 40% fewer parameters than BERT-base and runs about 60% faster while retaining about 97% of its language-understanding performance on the GLUE benchmark [14].

In MoodTune AI, DistilBERT is fine-tuned for six-class song emotion classification. The model processes the lyrics and predicts one of the predefined emotional categories. The predicted category is then used by the recommendation module to retrieve songs with the same emotion label.

The project also investigates approaches to address class imbalance, including resampling, class-weighted loss, and dataset expansion. These experiments suggest that improving emotion classification is not dependent on model architecture alone; dataset quality, label consistency, and the distribution of training examples are also important**.**

2.3 PROPOSED SYSTEM

MoodTune AI combines lyrics-based emotion classification with emotion-based music recommendation in a web application. The user enters a song title, and the application searches the local dataset for the corresponding lyrics. The lyrics are processed by a fine-tuned

DistilBERT model, which predicts one of six emotions: Anger, Fear, Happy, Love, Motivational, or Sad.

After classification, the application displays the predicted emotion and its model confidence score. It then selects up to five songs from the same emotional category and provides YouTube search links for access.

The final reported model achieves an accuracy of 67.6% and a Motivational-class F1-score of 0.52. These results indicate that the system can support emotion-aware song discovery, while also showing that further improvements in dataset quality, class representation, and model evaluation are needed.

The proposed system differs from conventional recommendation approaches by using the predicted emotional content of song lyrics as the main basis for selecting recommendations. It provides a foundation for future extensions such as personalized recommendations, improved emotion labeling, and audio-based emotion analysis.

CHAPTER 3. SYSTEM DESIGN

3.1 GENERAL

System design describes the architecture and major components of MoodTune AI. The system is designed to analyze song lyrics, predict their emotional category using a trained DistilBERT model, and recommend songs with matching emotion labels. It consists of a web interface, a Flask application, a machine learning model, and a local song dataset.

The system separates model training from application execution. During development, the model is trained and evaluated using prepared song lyrics. During application execution, the trained model is loaded to classify the lyrics of a selected song and generate recommendations.

3.2 SYSTEM ARCHITECTURE

The architecture of MoodTune AI consists of the following components:

1. User Interface: Allows users to enter a song title and view the predicted emotion, confidence score, and recommendations.
2. Flask Application: Receives user requests and coordinates the processing performed by the application.
3. Dataset Module: Searches the local song dataset and retrieves the lyrics and emotion information associated with songs.
4. DistilBERT Model: Processes the lyrics and predicts one of the six emotional categories.
5. Recommendation Module: Selects up to five songs belonging to the predicted emotional category.
6. YouTube Search Links: Provides links through which users can access the recommended songs.

The overall workflow begins when the user enters a song title. The application retrieves the corresponding lyrics, passes them to the trained model, and displays the prediction. The recommendation module then selects songs with the same emotion label.

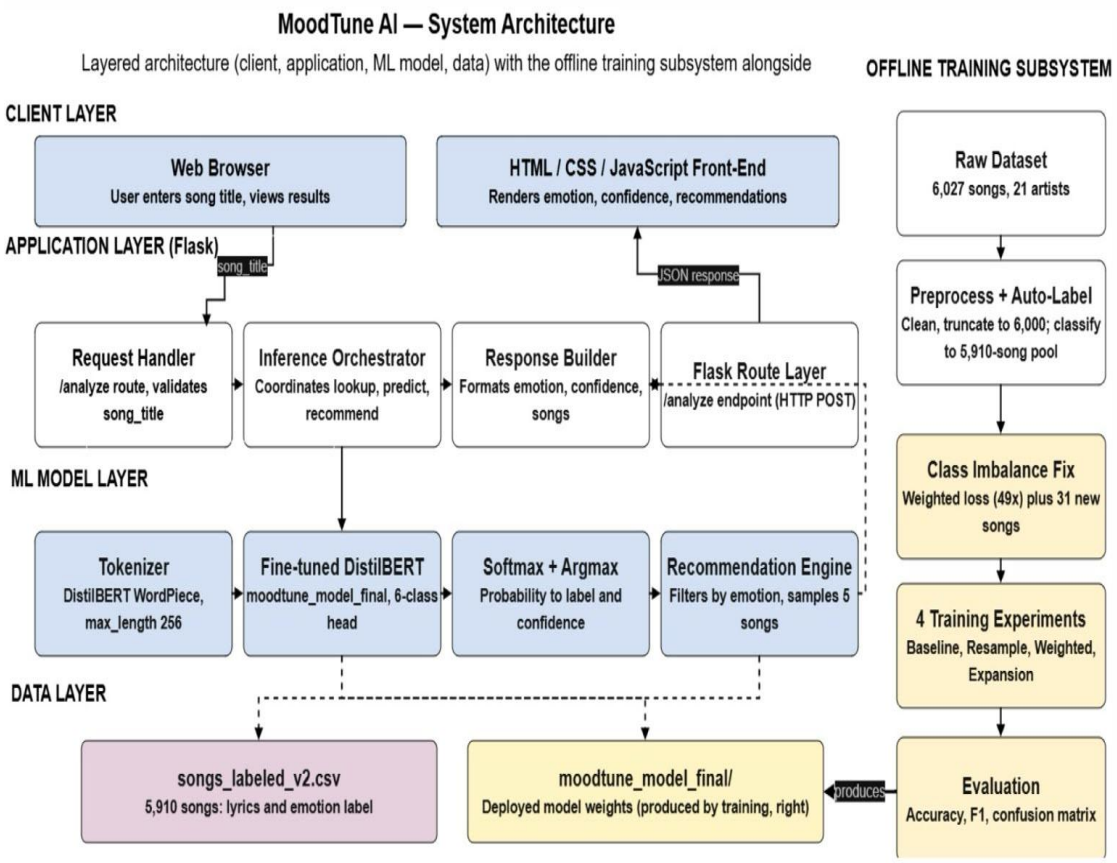


Figure 3.1: System Architecture of MoodTune AI

3.3 MODULE DESCRIPTION

MoodTune AI is divided into modules to simplify development, testing, and maintenance.

3.3.1 DATA COLLECTION MODULE

The data collection module gathers song titles and lyrics to create the dataset required for emotion classification. The collected data provides the textual input used to train and evaluate the model. The project uses a raw collection of 6,027 records.

3.3.2 DATA PREPROCESSING MODULE

The preprocessing module prepares the lyrics for model training. It organizes the song records, checks the available fields, and prepares the text and emotion labels for further processing. Consistent preprocessing helps reduce data quality problems and supports reliable model input.

3.3.3 AUTOMATED EMOTION LABELING MODULE

The automated labeling module assigns emotion categories to song lyrics using a pretrained text-classification approach. The resulting labels are used to prepare the dataset for supervised

fine-tuning. Since automatically generated labels may contain errors, label quality remains an important factor in the model's performance.

3.3.4 CLASS IMBALANCE HANDLING MODULE

The class imbalance handling module explores methods for improving the representation of less frequent emotional categories. The project investigates resampling, class-weighted loss, and dataset expansion. These approaches aim to improve the model's ability to recognize emotions that have fewer training examples.

3.3.5 MODEL TRAINING MODULE

The model training module uses DistilBERT for six-class emotion classification. The model is fine-tuned on the prepared lyrics dataset. The training process adjusts the model parameters to identify patterns associated with Anger, Fear, Happy, Love, Motivational, and Sad emotions.

3.3.6 MODEL EVALUATION MODULE

The evaluation module measures the classification performance using metrics such as accuracy, precision, recall, F1-score, and a confusion matrix. These measures help identify the overall performance of the model and reveal which emotion categories are classified more or less accurately.

3.3.7 WEB APPLICATION MODULE

The web application module is developed using Flask. It provides an interface through which users enter song titles and view the results. The backend coordinates dataset lookup, model inference, and recommendation generation.

3.3.8 RECOMMENDATION MODULE

The recommendation module uses the predicted emotion to select up to five songs with the same emotion label. It also provides YouTube search links for accessing the recommended songs. The current approach is emotion-based and does not use individual listening history or personal preferences.

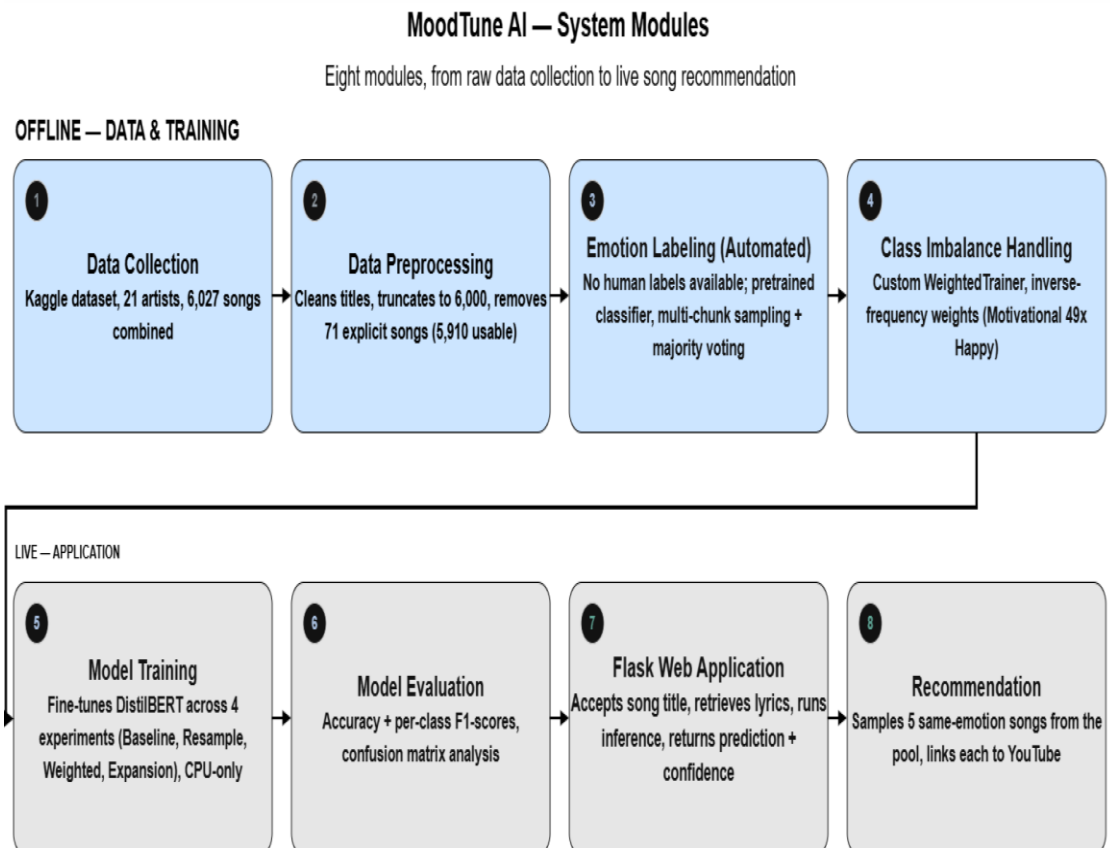


Figure 3.2: Modules of MoodTune AI

3.4 DATA FLOW

The data flow of MoodTune AI follows these steps:

1. The user enters a song title through the web interface.
2. The Flask application receives the request.
3. The dataset module searches for the song and retrieves its lyrics.
4. The lyrics are tokenized and processed using the DistilBERT tokenizer.
5. The trained model predicts the emotional category and produces class probabilities.
6. The application displays the predicted emotion and confidence score.
7. The recommendation module selects songs with the same emotion label and displays their YouTube search links.

If the requested song is not available in the local dataset, the application cannot perform the normal lyrics-based classification for that entry.

3.5 DATASET DESIGN

The dataset contains song records and emotion labels used for model development and recommendation.

Dataset	Number of Records
Raw song collection	6,027
Original labeled dataset	5,910
Expanded labeled dataset	5,941

The expanded dataset contains 31 additional Motivational examples compared with the original labeled dataset. The dataset is organized around six emotion categories: Anger, Fear, Happy, Love, Motivational, and Sad.

Dataset quality and class distribution are important because the model learns its classification patterns from the examples provided during training. Incorrect labels, duplicate records, and limited representation of certain emotions can affect the final results.

3.6 SOFTWARE REQUIREMENTS

The main software technologies used in the project are:

Software or Library	Purpose
Python	Main programming language
Flask	Web application backend
DistilBERT	Lyrics emotion classification
PyTorch	Deep learning model execution and training
Transformers	Tokenization and transformer model support
pandas	Dataset loading and manipulation
scikit-learn	Model evaluation and classification metrics
HTML and CSS	Web interface structure and styling

These technologies support the development of the machine learning pipeline and its integration into a web-based application.

3.7 DESIGN LIMITATIONS

The current design has several limitations. First, song lookup depends on the availability of lyrics in the local dataset. Second, automated emotion labels may contain errors, which can affect model training and predictions. Third, the model's reported accuracy indicates that some songs may be assigned an incorrect emotional category.

The confidence score is derived from the model's output probabilities and should not be interpreted as a guarantee that the prediction is correct. The recommendation module selects songs with matching emotion labels but does not personalise results using user preferences. Future improvements may include a larger and better-balanced dataset, manually verified labels, confidence calibration, and personalized recommendation techniques.

CHAPTER 4. IMPLEMENTATION

4.1 METHODOLOGY

The methodology describes the steps followed to develop MoodTune AI, from dataset preparation to the deployment of the emotion-based music recommendation application. The project combines Natural Language Processing, transformer-based emotion classification, model evaluation, and web application development.

The implementation is divided into data collection, data preprocessing, automated emotion labeling, class imbalance handling, model training, model evaluation, Flask application development, and song recommendation. Each stage contributes to the development of a system that predicts the emotional category of song lyrics and recommends songs with matching emotion labels.

4.2 DATA COLLECTION AND PREPROCESSING

The initial dataset contains 6,027 song records collected from 21 artists. The data is processed to prepare the song titles, lyrics, and emotion labels required for model development. During preprocessing, the collection is reduced to 5,910 usable records for the original labeled dataset.

The expanded labeled dataset contains 5,941 records, including 31 additional Motivational examples. The prepared dataset is used for training, evaluation, and emotion-based song retrieval.

The dataset counts in this report refer to three stages. The raw collection contains 6,027 song records from 21 artists. After cleaning and filtering, 5,910 usable records remain; this is the original labeled dataset. As summarized in Figures 3.1 and 3.2, this stage includes capping the collection at 6,000 records and removing 71 explicit songs. The expanded labeled dataset adds 31 Motivational examples, giving 5,941 records. The figure of 6,000 in the architecture diagram is therefore an intermediate cap and not a separate dataset.

4.2.1 CLASS DISTRIBUTION

The original labeled dataset contains six emotion categories: Happy, Anger, Sad, Fear, Love, and Motivational. The distribution is uneven, with Happy and Anger forming the largest classes and Motivational having the fewest examples. Figure 4.1 shows this distribution.

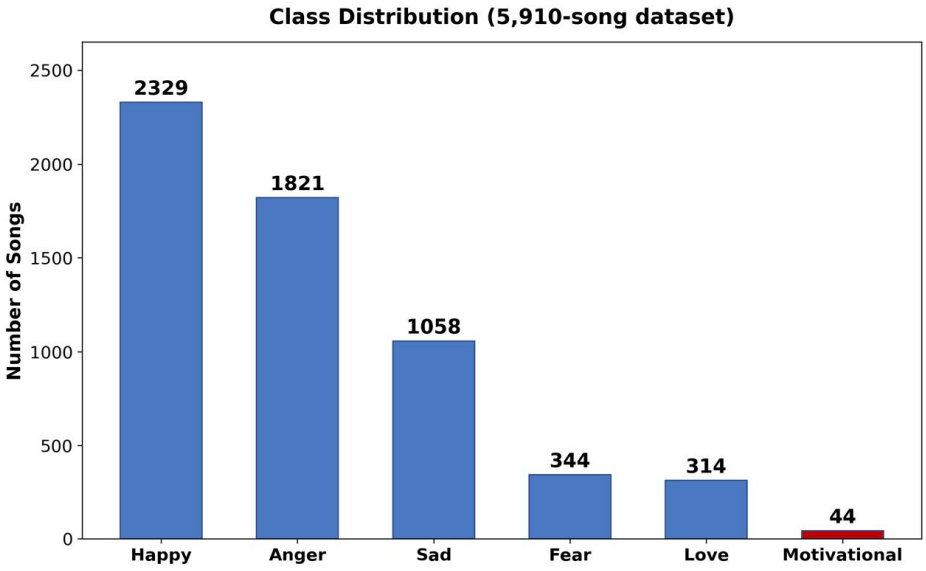


Figure 4.1: Distribution of Emotion Classes in the Dataset

The class distribution is important because a model trained on an imbalanced dataset may favor categories with more examples. In this dataset, the Motivational class contains only 44 records, compared with 2,329 Happy records. This imbalance presents a challenge for reliable classification across all six emotions.

4.3 AUTOMATED EMOTION LABELING

Emotion labels are required to train the DistilBERT model as a supervised classification system. The project uses an automated labeling approach to assign emotional categories to song lyrics. The labeling process uses a pretrained text classifier and combines predictions from multiple text chunks through majority voting.

Automated labeling reduces the need to manually annotate every song, but it can introduce label noise. Some lyrics express more than one emotion, and the emotional meaning of a song may not always be represented accurately by a single category. Therefore, the quality of the assigned labels can affect the performance of the final model.

4.4 CLASS IMBALANCE HANDLING

The dataset contains substantially different numbers of examples for each emotional category. The project investigates several methods to improve the classification of underrepresented classes.

The approaches evaluated include baseline training without class weighting, resampling, class-weighted loss, and dataset expansion. The weighted approach assigns greater importance to

less frequent categories during training. Dataset expansion adds 31 Motivational examples to the original labeled dataset.

These experiments help assess the trade-off between overall classification accuracy and performance on the Motivational category.

4.5 MODEL TRAINING

MoodTune AI uses DistilBERT, a transformer-based language model, for song lyric emotion classification. The model is fine-tuned using the prepared dataset and configured to predict one of six emotional categories: Anger, Fear, Happy, Love, Motivational, and Sad.

The lyrics are converted into token representations using the DistilBERT tokenizer. The input sequence is configured with a maximum length of 256 tokens. The model produces scores for the six emotion classes, which are converted into probabilities for prediction.

Four training experiments are considered: baseline training, resampling, class-weighted loss, and dataset expansion. The reported results are compared to understand how the approaches affect overall accuracy and the classification of the Motivational category.

4.6 MODEL EVALUATION

The model is evaluated using accuracy, class-wise F1-scores, and a confusion matrix. Accuracy measures the proportion of correctly classified examples, while the F1-score provides a balance between precision and recall for a particular class. The confusion matrix shows the relationship between actual and predicted emotion categories. Figure 4.2 summarizes the overall accuracy and the Motivational F1-score of the four experiments, and Figure 4.3 compares class-wise F1-scores.

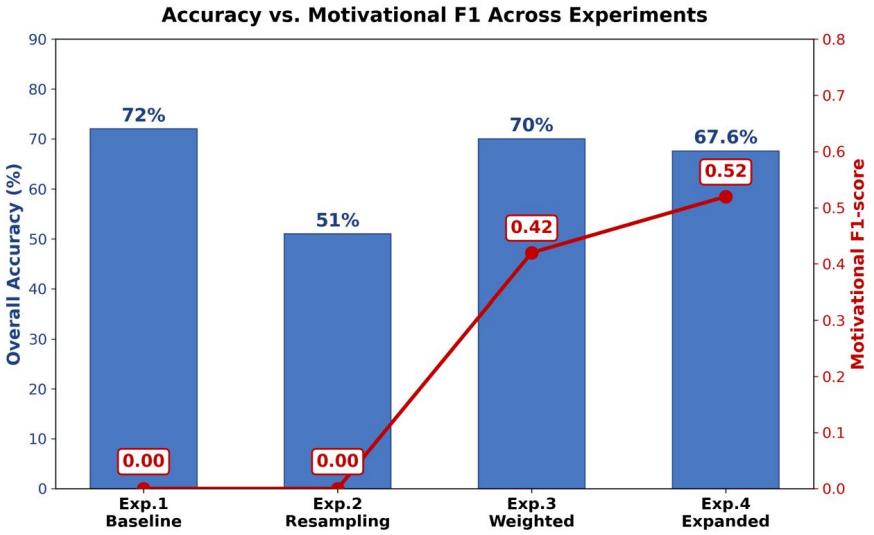


Figure 4.2: Comparison of Accuracy and Motivational F1-Score Across Experiments

The reported results show that the baseline experiment achieved 72% accuracy but a Motivational F1-score of 0.00. Resampling achieved 51% accuracy and a Motivational F1-score of 0.00. Class-weighted loss achieved 70% accuracy and a Motivational F1-score of 0.42. Dataset expansion achieved 67.6% accuracy and a Motivational F1-score of 0.52.

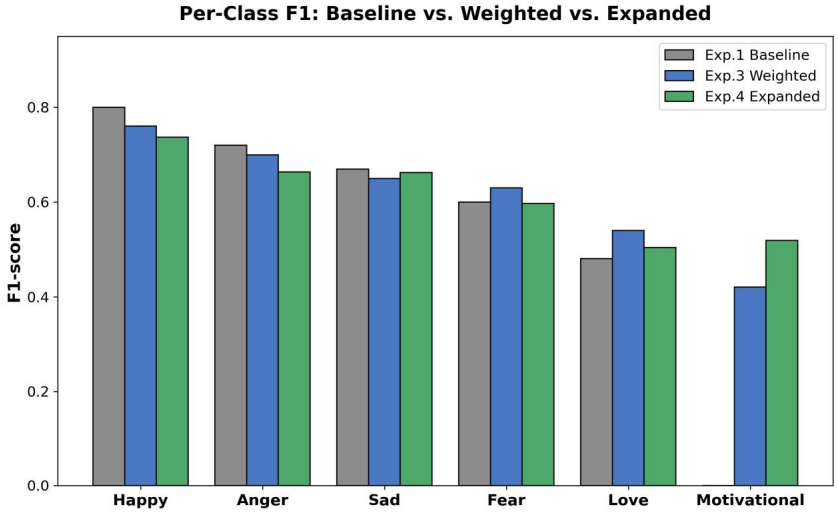


Figure 4.3: Per-Class F1-Score Comparison

Figure 4.3 does not include Experiment 2 (resampling), whose accuracy was 51% and whose Motivational F1-score was 0.00. These results indicate that dataset expansion improved the reported Motivational F1-score, although the final model's overall accuracy was lower than the baseline accuracy.

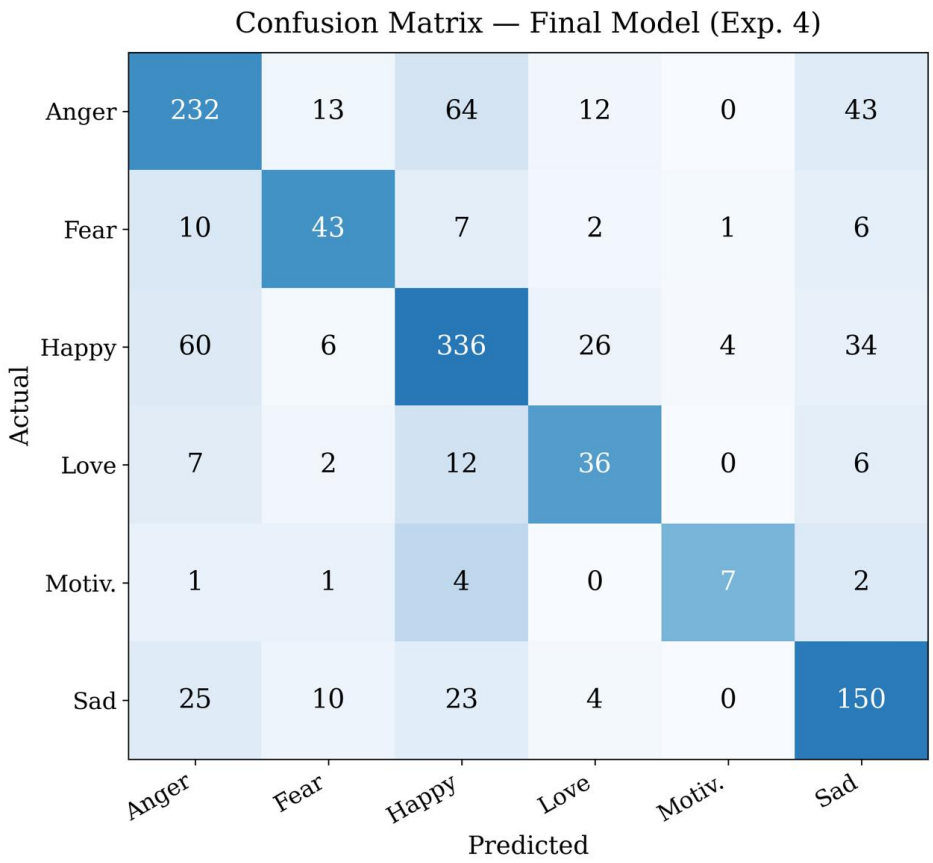


Figure 4.4: Confusion Matrix of the Final DistilBERT Model

The confusion matrix in Figure 4.4 illustrates the classification performance of the final model across the six emotion categories. The model correctly classified 804 out of 1,189 test examples, resulting in an accuracy of approximately 67.6%.

The matrix shows that Anger was frequently confused with Happy and Sad, with 64 and 43 examples respectively. Similarly, 60 Happy examples were predicted as Anger. These errors indicate that some song lyrics contain overlapping emotional expressions, making accurate classification difficult.

The Motivational category had seven correctly classified examples out of fifteen actual examples in the displayed matrix. This highlights the difficulty of classifying underrepresented emotions. Dataset imbalance, automated labeling errors, and similarities between emotional expressions may contribute to classification errors.

Overall, the confusion matrix identifies the classes in which the model makes errors and provides useful information for future improvements.

4.7 DEMONSTRATION OF THE WEB APPLICATION

The MoodTune AI web application was developed using Flask and provides a simple interface for analysing the emotional content of song lyrics. The user enters a song title in the input field and clicks the Analyze button. The application retrieves the corresponding lyrics from the dataset and passes them to the fine-tuned DistilBERT model for emotion classification.

The predicted emotion and its confidence score are displayed on the results page. The application also recommends up to five songs belonging to the same emotion category. Each recommendation includes a link that opens a YouTube search for the song.

The application was demonstrated using songs representing all six emotion categories: Anger, Fear, Happy, Love, Motivational, and Sad. The following screenshots illustrate the output produced by the application for each category (Figures 4.5 to 4.10).

ANGER EMOTION

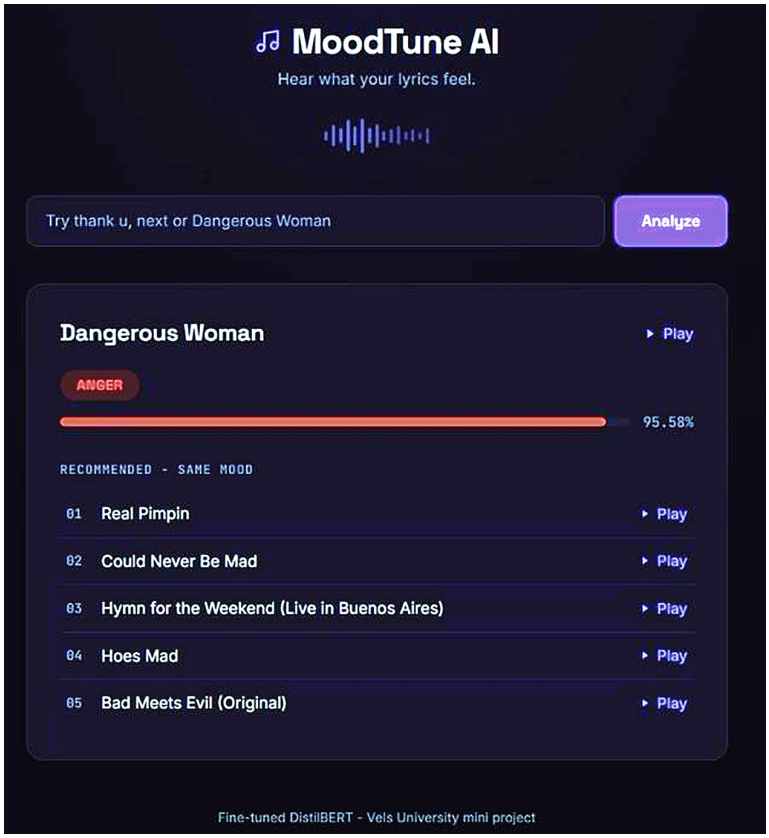


Figure 4.5: Web Application Output for the Anger Example (“Dangerous Woman”)

FEAR EMOTION

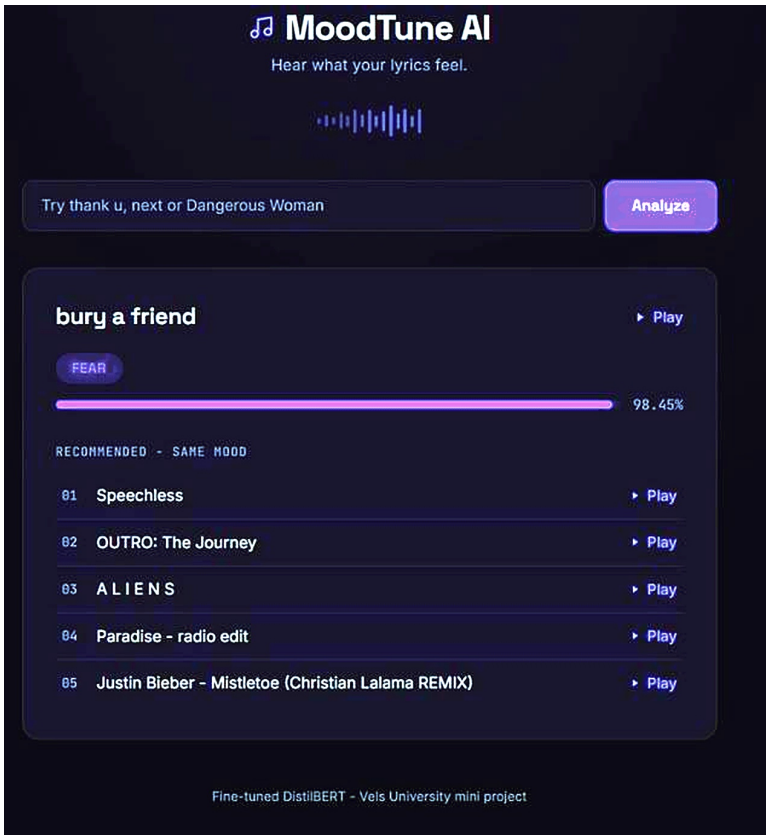


Figure 4.6: Web Application Output for the Fear Example (“bury a friend”)

HAPPY EMOTION

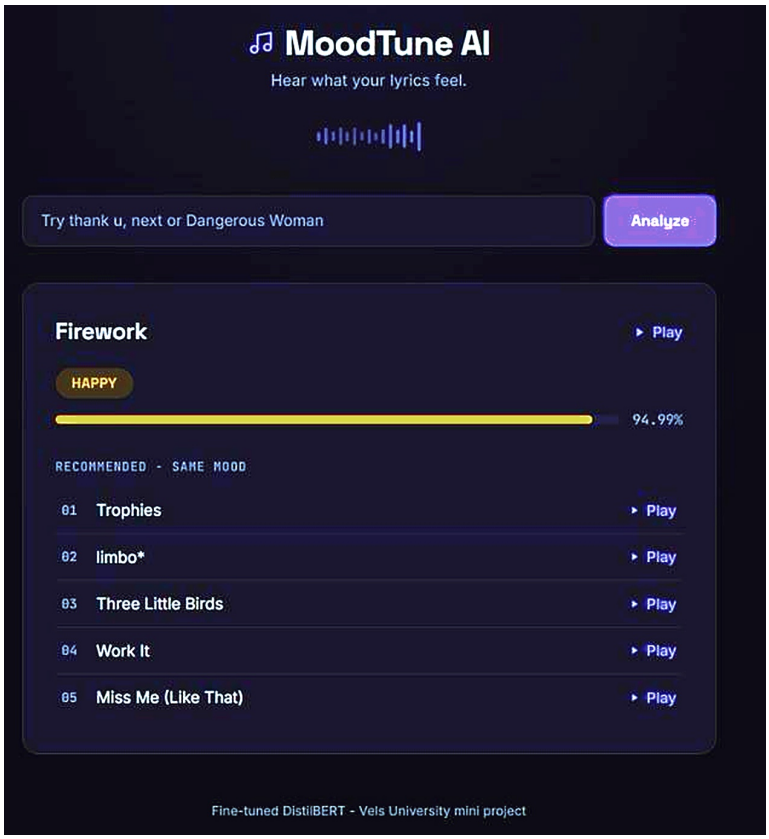


Figure 4.7: Web Application Output for the Happy Example (“Firework”)

LOVE EMOTION

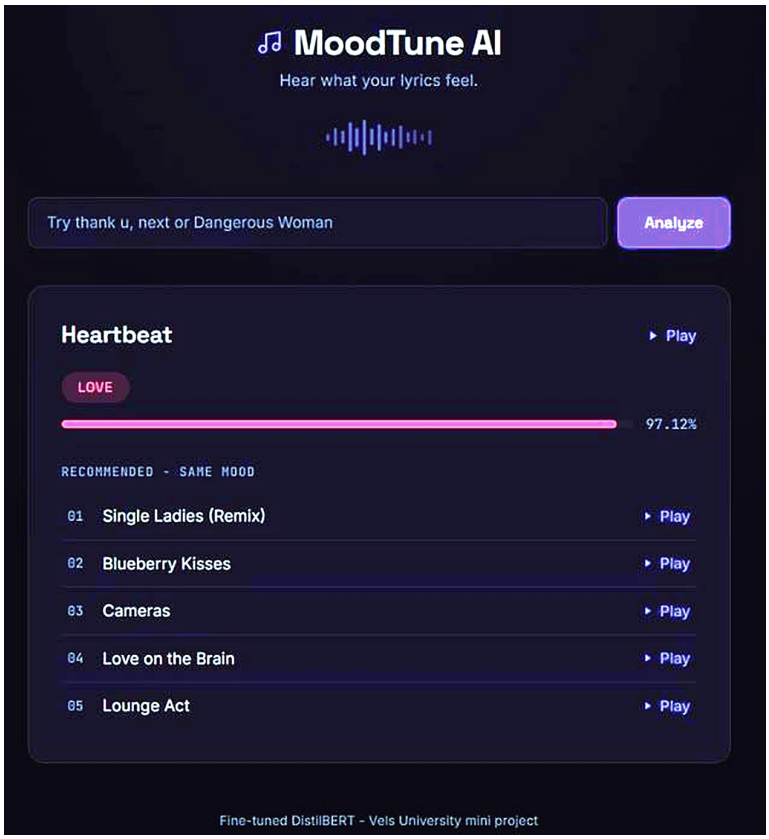


Figure 4.8: Web Application Output for the Love Example (“Heartbeat”)

MOTIVATIONAL EMOTION

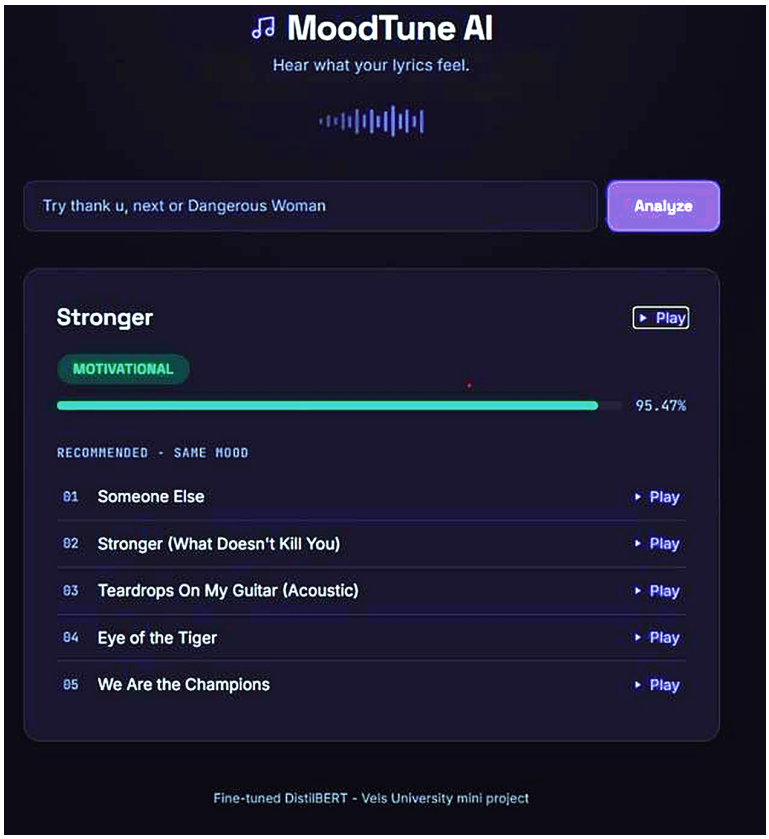


Figure 4.9: Web Application Output for the Motivational Example (“Stronger”)

SAD EMOTION

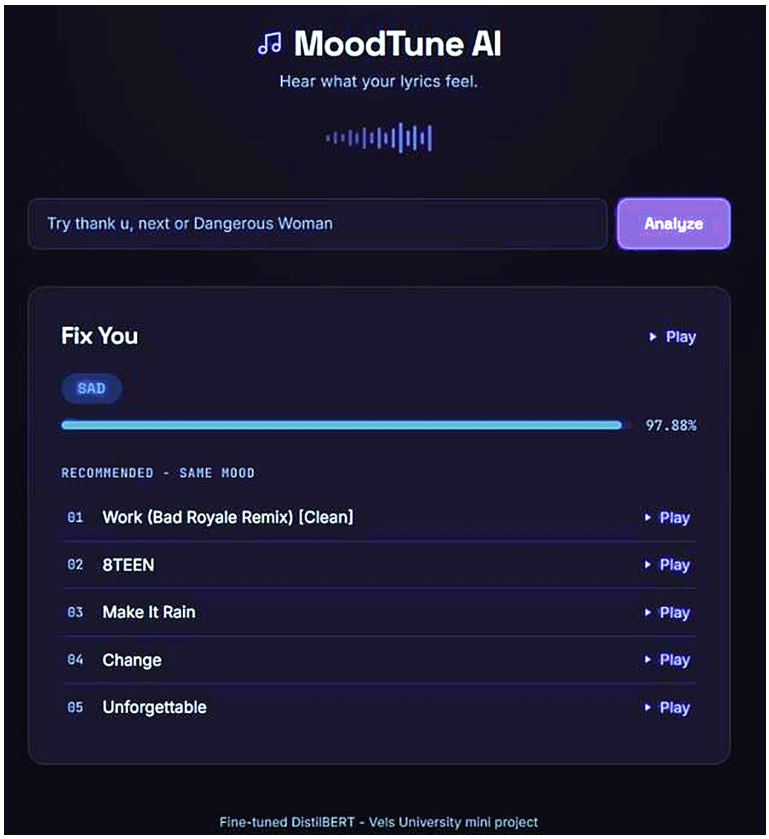


Figure 4.10: Web Application Output for the Sad Example (“Fix You”)

Demonstration Summary

The screenshots demonstrate that the application can display predicted emotions, confidence scores, and same-emotion song recommendations through a consistent web interface. The six examples cover all emotion categories supported by the system.

4.8 IMPORTANT CODE IMPLEMENTATION

This section presents the important code components of MoodTune AI. The implementation integrates the Flask web framework, the fine-tuned DistilBERT model, confidence score calculation, and an emotion-based music recommendation module. These components work together to analyze song lyrics, predict the corresponding emotion, and recommend songs belonging to the same emotional category.

4.8.1 FLASK REQUEST HANDLING

The Flask application acts as the interface between the user and the emotion analysis module. It receives the song title submitted through the web form, passes it to the analyze_song() function, and displays the resulting information on the web page.

Code snippet:

```python
from flask import Flask, render_template, request
from app_logic import analyze_song

app = Flask(__name__)

@app.route("/", methods=["GET", "POST"])
def home():
    result = None
    if request.method == "POST":
        song_name = request.form["song_name"]
        result = analyze_song(song_name)
    return render_template("index.html", result=result)

if __name__ == "__main__":
    app.run(debug=True)
```

Explanation:

The Flask request-handling module provides the interface between the user and the MoodTune AI system. It accepts the song name submitted through the web form, calls the analyze_song() function from app_logic.py, and passes the analysis result to the HTML template for display. The application supports both GET and POST requests. The snippet runs Flask with debug mode enabled, which is suitable for development only.

4.8.2 DISTILBERT EMOTION PREDICTION

```python
import pandas as pd
import torch
from transformers import DistilBertTokenizer, DistilBertForSequenceClassification
import torch.nn.functional as F

# Load your trained model
tokenizer = DistilBertTokenizer.from_pretrained("moodtune_model_final")
model = DistilBertForSequenceClassification.from_pretrained("moodtune_model_final")
model.eval()

# Load your labeled dataset (for lookup + recommendations)
df = pd.read_csv("data/songs_labeled_v2.csv")
df["Title"] = df["Title"].str.replace("\u200b", "", regex=False).str.strip()

id2label = model.config.id2label

def get_lyrics_by_song_name(song_name):
    match = df[df["Title"].str.lower() == song_name.lower()]
    if match.empty:
        return None
    return match.iloc[0]["Lyric"]

def predict_emotion(lyrics):
    inputs = tokenizer(str(lyrics), return_tensors="pt", truncation=True, padding=True, max_length=256)
    with torch.no_grad():
        outputs = model(**inputs)
    probs = F.softmax(outputs.logits, dim=1)
    confidence, predicted_class = torch.max(probs, dim=1)
    emotion = id2label[predicted_class.item()]
    return emotion, round(confidence.item() * 100, 2)
```

The emotion prediction module uses the trained DistilBERT model to identify the emotional category of a song from its lyrics. The lyrics are converted into tokens using the tokenizer, with a maximum sequence length of 256. The model processes these tokens to generate logits, which are converted into probabilities using the softmax function. The class with the highest probability is selected as the predicted emotion, and its probability is returned as the confidence score.

The confidence score is the softmax probability of the predicted class. It shows how strongly the model favors that class compared with the other five, but it is not a guarantee that the prediction is correct: the model can assign a high confidence to an incorrect emotion, and the scores were not calibrated in this project. Lyrics longer than 256 tokens are truncated, so only the first part of a long song is analyzed. The lookup matches the song title exactly, ignoring case, and uses the first matching record.

4.8.3 EMOTION-BASED SONG RECOMMENDATION

```python
def recommend_songs(emotion, exclude_song=None, top_n=5):
    matches = df[df["emotion"] == emotion]
    if exclude_song:
        matches = matches[matches["Title"].str.lower() != exclude_song.lower()]
    return matches["Title"].sample(n=min(top_n, len(matches)), random_state=None).tolist()

def analyze_song(song_name):
    lyrics = get_lyrics_by_song_name(song_name)
    if lyrics is None:
        return {"error": "Song not found in dataset"}
    emotion, confidence = predict_emotion(lyrics)
    recommendations = recommend_songs(emotion, exclude_song=song_name)
    recommendations_with_links = [
        {
            "title": song,
            "youtube_search": song.replace(' ', '+'),
        }
        for song in recommendations
    ]
    return {
        "song": song_name,
        "emotion": emotion,
        "confidence": confidence,
        "recommendations": recommendations_with_links
    }
    # Quick test
if __name__ == "__main__":
    result = analyze_song("thank u, next")
    print(result)
```

The recommendation module selects songs that belong to the predicted emotional category. It filters the dataset using the detected emotion and excludes the input song from the recommendations. Up to five songs are randomly selected from the matching results. The system then prepares the song titles and YouTube search information for display on the web page.

The recommended songs are selected using the emotion label stored in the dataset, which was assigned by the automated labeling process, and not by running the model on every candidate song. The YouTube output is a search query built from the song title.

4.9 CHAPTER SUMMARY

This chapter presented the methodology, implementation, and evaluation of the MoodTune AI system. It described the dataset collection and preprocessing steps, automated emotion labeling, class imbalance handling, and DistilBERT model training. The model was evaluated using accuracy, class-wise F1-scores, and a confusion matrix. The web application demonstration showed how users can enter a song title, view the predicted emotion and confidence score, and receive recommendations for songs belonging to the same emotional category. The implementation details of Flask request handling, emotion prediction, and emotion-based song recommendation were also presented.

CHAPTER 5. RESULTS AND DISCUSSION

5.1 INTRODUCTION

This chapter presents the results obtained from the MoodTune AI system and discusses the performance of the DistilBERT-based song emotion classification model. The system classifies songs into six emotional categories: Anger, Fear, Happy, Love, Motivational, and Sad. The performance is analyzed using classification accuracy, class-wise F1-scores, and the confusion matrix. The chapter also discusses the experimental approaches used to improve the recognition of underrepresented emotional categories and the results obtained from the web application demonstration.

5.2 MODEL PERFORMANCE EVALUATION

The performance of the MoodTune AI system was evaluated using classification accuracy, class-wise F1-scores, and a confusion matrix. The final DistilBERT model classifies songs into six emotional categories: Anger, Fear, Happy, Love, Motivational, and Sad.

The final evaluation achieved an accuracy of approximately 67.6% on the test dataset. The results indicate that the model can identify several emotional categories, although its performance varies across classes. The Motivational category remains challenging because it contains fewer labeled examples than the other categories.

The confusion matrix provides further information about correct predictions and misclassifications. These results help identify the strengths and limitations of the model and provide a basis for future improvements.

5.3 COMPARISON OF EXPERIMENTAL APPROACHES

Several experimental approaches were evaluated to improve the emotion classification performance of the MoodTune AI system. These experiments focused on improving the recognition of the underrepresented Motivational emotion category while maintaining overall classification performance.

The baseline model achieved 72% accuracy, but its F1-score for the Motivational category was 0.00. A resampling approach resulted in 51% accuracy and did not improve the Motivational F1-score. The class-weighted loss approach achieved 70% accuracy and improved the Motivational F1-score to 0.42.

Finally, the dataset was expanded by adding 31 additional Motivational examples. This approach achieved 67.6% accuracy and a Motivational F1-score of 0.52. Although the final model had lower overall accuracy than the baseline, it demonstrated improved recognition of the Motivational category.

These experiments show a trade-off between overall accuracy and performance on a minority class. Overall accuracy is dominated by the large classes, so a model can obtain a high accuracy while ignoring a small class: the baseline reached 72% accuracy but an F1-score of 0.00 for Motivational. The final model has a lower accuracy (67.6%) but a Motivational F1-score of 0.52, so it is not worse in every respect; whether the trade is worthwhile depends on how much recognizing the Motivational class matters. The comparison should be read with care: the Motivational class has only 15 of the 1,189 test samples, so a change in F1-score from 0.42 to 0.52 corresponds to a small number of predictions.

5.4 CONFUSION MATRIX AND CLASS-WISE RESULTS

The confusion matrix was used to examine the prediction performance of the final MoodTune AI model across the six emotion categories. It compares the actual emotion labels with the emotions predicted by the model. The diagonal entries represent correct predictions, while the off-diagonal entries represent misclassifications.

The model correctly classified 804 out of 1,189 test samples, resulting in an accuracy of approximately 67.6%. The results show that the model performs differently across the emotion categories. The Happy category had 336 correct predictions, while Anger had 232 and Sad had 150 correct predictions.

The Motivational category had 7 correct predictions out of 15 samples. This indicates that the model still has difficulty recognizing this underrepresented category. Some Anger samples were predicted as Happy or Sad, while some Happy samples were predicted as Anger. These errors show that the model sometimes confuses emotional categories with overlapping characteristics.

Overall, the confusion matrix provides a detailed view of the model's strengths and weaknesses. It also indicates that further improvements in dataset quality, class representation, and model training may help improve classification performance.

5.5 WEB APPLICATION RESULTS

The MoodTune AI web application provides a simple interface for song emotion analysis and emotion-based music recommendation. Users can enter a song title, and the application searches the available dataset for the corresponding lyrics. If the song is found, the lyrics are processed by the trained DistilBERT model to predict the emotional category.

The application displays the predicted emotion along with its confidence score. It also provides recommendations for up to five songs belonging to the same predicted emotion. YouTube search links are provided to help users find the recommended songs.

During the demonstration, the application produced predictions for songs representing all six emotion categories: Anger, Fear, Happy, Love, Motivational, and Sad. These examples demonstrated the complete workflow from song-title input to emotion prediction and recommendation.

The web application connects the trained model with a user-friendly interface, making the system accessible without requiring users to interact directly with the machine-learning code. However, the results depend on the availability and quality of the lyrics in the dataset, and the confidence score should not be interpreted as a guarantee that the prediction is correct.

5.6 LIMITATIONS OF THE RESULTS

Although the MoodTune AI system successfully demonstrates song emotion classification and emotion-based music recommendation, several limitations remain.

First, the model's overall classification accuracy is approximately 67.6%, indicating that some songs are incorrectly classified. The confusion matrix also shows that certain emotion categories are more difficult to distinguish than others.

Second, the Motivational category has fewer test samples than the other categories. Although dataset expansion improved its F1-score from 0.42 to 0.52, its recognition performance still requires improvement. The quality of automatically assigned emotion labels may also affect model performance.

Third, the system depends on the availability of song lyrics in its dataset. If a song is not found, the application may be unable to analyze it. In addition, the model analyzes lyrics rather than the actual audio, so it does not directly consider musical features such as tempo, melody, rhythm, or vocal expression.

Finally, the recommendation module selects songs from the same predicted emotion category. It does not learn individual user preferences or provide personalized recommendations. These limitations identify areas for future development.

5.7 CHAPTER SUMMARY

This chapter discussed the results and performance of the MoodTune AI system. The final DistilBERT model achieved an overall classification accuracy of approximately 67.6%. Experimental comparisons showed that class-weighted loss and dataset expansion improved the recognition of the underrepresented Motivational category. The confusion matrix provided insights into correct predictions and misclassifications across the six emotion categories.

The web application successfully demonstrated the workflow of song-title input, lyric-based emotion prediction, confidence-score display, and emotion-based song recommendation. The chapter also identified limitations related to classification accuracy, dataset quality, limited representation of certain emotions, and the absence of audio-based analysis and personalized recommendations. These findings provide a foundation for the conclusions and future enhancements discussed in the next chapter.

CHAPTER 6. CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

MoodTune AI was developed as a song emotion analysis and emotion-based music recommendation system using DistilBERT. The system classifies song lyrics into six emotional categories: Anger, Fear, Happy, Love, Motivational, and Sad. It integrates a trained transformer model with a Flask web application to provide an accessible interface for song analysis.

The final model achieved an overall classification accuracy of approximately 67.6%. Experimental comparisons suggest that class-weighted loss and dataset expansion improved the recognition of the underrepresented Motivational category. The web application demonstrates the complete process of entering a song title, predicting its emotion, displaying a confidence score, and recommending songs from the same emotional category.

Although the system provides a useful demonstration of transformer-based text classification, its performance is affected by dataset quality, class imbalance, and confusion between some emotional categories. Overall, the project demonstrates the practical application of natural language processing and deep learning to lyric-based emotion classification and music recommendation. With an accuracy of about 67.6%, roughly one in three test songs was assigned a different emotion from its label, so the system should be regarded as a prototype and not as a reliable emotion detector.

6.2 FUTURE ENHANCEMENTS

The MoodTune AI system can be improved in several ways to enhance its accuracy, functionality, and user experience.

1. Dataset Expansion: A larger and more balanced dataset can be created by collecting additional song lyrics for all six emotional categories, particularly the Motivational category.
2. Improved Emotion Labeling: More reliable labeling methods and manual verification can help reduce incorrect emotion labels and improve training quality.
3. Model Optimization: Further experiments with hyperparameters, class-weighted training, and suitable transformer models may improve overall accuracy and class-wise F1-scores.
4. Audio-Based Emotion Analysis: Future versions can analyze audio features such as tempo, rhythm, melody, and vocal expression alongside lyrics to provide a more comprehensive understanding of song emotion.

5. Personalized Recommendations: User preferences, listening history, and feedback can be incorporated to recommend songs that better match individual interests [12].
6. Enhanced Web Application: The application can be extended with a larger song database, improved search functionality, user accounts, and a more interactive interface.
7. Real-Time Music Integration: Integration with music streaming platforms, where permitted by their APIs and terms, could provide easier access to songs and recommendations.

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